



Studybeesofficial

Where Minds Pollinate

Class 10

Biology

Chapter 1

Notes+Short Questions

+Long Questions

NBF book

Follow Our Socials



studybeesofficial



studybeesofficialpak



studybeesofficialpak

BIOLOGY

CLASS: 10TH

CHAPTER: 1 – DIGESTIVE SYSTEM

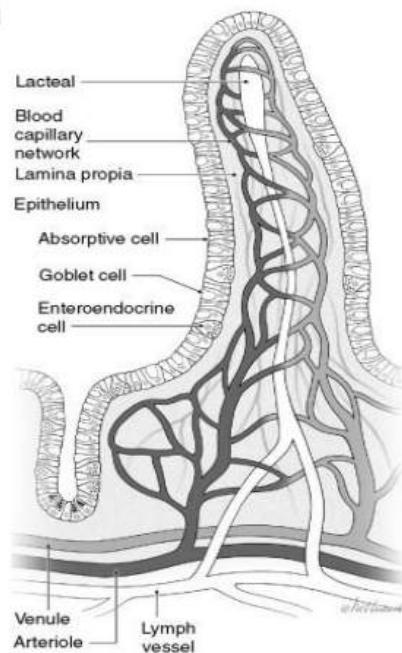
SHORT QUESTIONS:

1. Write the functions of: saliva, gall bladder, pancreas, villus, bile, gastric gland, renin, HCl in stomach, hepatic portal vein.

Answer:

- **Saliva:** Moistens food, making it easier to chew and swallow. It contains salivary amylase that starts digestion of starch.
- **Gall bladder:** Stores and concentrates bile, and releases it into the duodenum to help in fat digestion.
- **Pancreas:** Secretes digestive enzymes (amylase, lipase, proteases) and hormones like insulin.
- **Villus:** Increases the surface area in the small intestine to absorb nutrients efficiently.
- **Bile:** Emulsifies fats into small droplets for easier digestion by lipase.
- **Gastric gland:** Produces gastric juice containing HCl, pepsinogen, and mucus for protein digestion.
- **Renin:** An enzyme in infants that helps digest milk by clotting milk proteins.
- **HCl in stomach:** Kills bacteria, provides an acidic environment for enzyme activation.
- **Hepatic portal vein:** Carries absorbed nutrients from the small intestine to the liver for processing.

2. Draw and label longitudinal section of villus.



3. Name the structures that secrete enzymes necessary for human digestion.

Answer:

The following structures secrete enzymes:

- **Salivary glands** (produce salivary amylase)
- **Gastric glands in stomach** (produce pepsinogen and HCl)
- **Pancreas** (produces amylase, lipase, and proteases)
- **Intestinal glands** (produce enzymes like maltase, sucrase, peptidase)

4. Why is pepsin formed in inactive form pepsinogen?

Answer:

Pepsin is a strong protein-digesting enzyme that can damage the stomach lining if active. It is secreted as an inactive form called **pepsinogen**, which is then activated by HCl to prevent damage to stomach tissues.

5. Why must digested food be distributed by blood in man and other animals?

Answer:

Digested food contains nutrients needed by all body cells. Blood distributes these nutrients from the intestines to every part of the body so they can be used for energy, growth, and repair.

6. How is chewing important to your digestion?

Answer:

Chewing physically breaks food into smaller pieces, increasing its surface area. It mixes food with saliva, which begins starch digestion, and forms a soft bolus that is easy to swallow.

7. What would happen if pancreatic juice did not reach your small intestine?

Answer:

Pancreatic juice contains essential enzymes like amylase, lipase, and proteases. If it doesn't reach the small intestine, carbohydrates, proteins, and fats would not be digested properly, leading to malnutrition and digestive issues.

8. Why pepsin is unable to perform its enzymatic activity in small intestine?

Answer:

Pepsin works in an acidic environment like the stomach. The small intestine is alkaline due to the presence of bile and pancreatic juice, which makes pepsin inactive and unable to digest proteins there.

9. Which parts of digestive system can come in contact with bile?

Answer:

Bile is secreted by the liver and stored in the gall bladder. It is released into the **duodenum**, the first part of the small intestine. So, only the **small intestine** comes in contact with bile.

10. Some cells of the stomach secrete mucous for protection of stomach wall. Suggest why the stomach wall needs to be protected against it.

Answer:

The stomach produces strong acid (HCl) and enzymes (like pepsin) that can digest proteins. Without mucus, these secretions would damage the stomach wall, causing ulcers or wounds.

11. Complete the table by writing names of substrate and product for each enzyme.

Answer:

Name of Enzyme	Substrate	Product
Protease	Proteins	Amino acids
Amylase	Starch	Maltose or Glucose
Lipase	Lipids (fats)	Fatty acids + Glycerol

12. Write the function of each type of teeth.

Answer:

- **Incisors:** Sharp front teeth used for cutting food.
 - **Canines:** Pointed teeth next to incisors, used for tearing food.
 - **Premolars:** Flat-surfaced teeth for crushing and grinding food.
 - **Molars:** Larger teeth at the back of the mouth used for grinding and chewing food thoroughly.
-

13. Describe digestion in oral cavity.

Answer:

Digestion begins in the oral cavity where food is chewed by teeth and mixed with saliva. Saliva contains **salivary amylase**, which starts the chemical digestion of starch into simpler sugars. The tongue helps in mixing and forming the bolus.

14. Differentiate between mechanical and chemical digestion.

Answer:

Mechanical Digestion	Chemical Digestion
Physical breakdown of food (e.g., chewing, churning)	Chemical breakdown by enzymes (e.g., amylase, pepsin, lipase)
Increases surface area for enzymes	Converts complex food into absorbable molecules
Takes place in mouth and stomach	Takes place in mouth, stomach, and small intestine

CH# 1 DIGESTIVE SYSTEM

SLO QUESTION AND ANSWER

Q1: Why doesn't digestion occur inside the esophagus?

The esophagus completely lacks any digestive glands, goblet secretory tissues, or enzyme-producing cells. Its internal lining is anatomically designed strictly for mucous lubrication and mechanical transport of food. Its sole functional purpose is to safely move the food bolus down to the stomach using rhythmic waves called peristalsis.

Q2: What would happen to the movement of food if the circular muscles of the gut wall fail to contract?

The coordinated process of peristalsis will stop completely because waves cannot propagate along the digestive tract. The food bolus will become entirely stuck and stagnant at that specific zone of the digestive canal. The gut wall cannot contract directly behind the food to narrow the tube and push the bolus forward.

Q3: How does the structure of the human tongue assist both respiratory and digestive systems?

In digestion, it moves food around the mouth to facilitate chewing and rolls it into a soft bolus for swallowing. **In respiration**, it helps pull the larynx upward and forward during the automatic mechanism of swallowing food. **In protection**: This upward movement forces the epiglottis to firmly seal the airway, preventing food from entering the windpipe.

Q4: Differentiate between the functions of a cardiac sphincter and a pyloric sphincter.

- **Cardiac Sphincter**: It acts as an entry valve that regulates the passage of swallowed food from the esophagus into the stomach.
- **Cardiac Sphincter**: It prevents highly corrosive and acidic gastric juices from leaking backward into the delicate esophagus.
- **Pyloric Sphincter**: It acts as an exit valve that controls the slow, gradual release of acidic chyme into the duodenum.

Q5: Why do carnivores or humans have highly developed canine teeth compared to herbivores?

Canines are structurally sharp, pointed, and dagger-like teeth situated at the corners of the dental arch. They are specifically adapted for piercing, gripping, and tearing tough

animal muscle fibers and raw flesh apart. Herbivores live entirely on plant matter and only require flat, wide teeth to grind down tough cellulose cell walls.

Q6: Salivary amylase works perfectly in the mouth but becomes completely inactive in the stomach. Why?

Salivary amylase is a highly pH-sensitive enzyme that requires a neutral to slightly alkaline environment to stay active. The stomach lining secretes highly concentrated hydrochloric acid, creating a severely acidic environment with a very low pH. This intense acidity completely denatures the delicate protein structure of salivary amylase, permanently halting its catalytic action.

Q7: If a person's bile duct is blocked by a gallstone, which nutrient group will they struggle to digest?

The patient will severely struggle to digest all dietary fats, lipids, and fat-soluble vitamins. Bile contains no chemical digestive enzymes but is essential for the physical process of fat emulsification. Without bile, large fat globules cannot be broken down into microscopic droplets, leaving very little surface area for lipase enzymes.

Q8: Why is Enterokinase called an activating enzyme rather than a digestive enzyme?

Enterokinase does not break down any food particles, nutrient molecules, or chemical peptide bonds directly. Its single, specific regulatory function is to chemically activate dormant enzyme precursors secreted by the pancreas. It converts the inactive trypsinogen into its highly potent, active protein-digesting form called trypsin inside the intestine.

Q9: Infants produce the enzyme Rennin, but adults do not. What is the physiological reason behind this?

Human infants rely entirely on a liquid milk diet, whereas adults consume a variety of solid foods. Rennin is specifically designed to curdle and solidify soluble milk proteins like casein inside the infant's stomach. Turning liquid milk into a solid curd slows its passage, ensuring it stays in the gut long enough for complete digestion.

Q10: Why are the secretions of the pancreas and liver alkaline, whereas stomach secretions are highly acidic?

Stomach acid is required to provide a low pH to activate pepsin and kill harmful microbes entering with food. Conversely, the digestive enzymes produced by the pancreas and small intestine can only function in a high pH alkaline medium. Alkaline bile and pancreatic secretions neutralize the acidic chyme to protect duodenal tissues and activate intestinal enzymes.

Q11: Why do amino acids and glucose enter the blood capillaries of a villus, while fatty acids enter the lacteal?

Glucose and amino acids are water-soluble molecules that easily diffuse across membranes directly into blood capillaries. Fatty acids and glycerol are lipid-soluble and recombine into large fat particles inside the intestinal epithelial cells. These fat aggregates are too large to pass through tight capillary walls, so they enter the wider, highly permeable lymphatic lacteals.

Q12: What is the biological significance of the Hepatic Portal Vein?

It ensures that all absorbed nutrients, toxins, and chemicals go directly from the gut to the liver before entering general circulation. This unique pathway allows the liver to filter out, neutralize, and destroy harmful substances or toxins immediately. It enables the liver to monitor blood content, store excess glucose as glycogen, and process vital vitamins right away.

Q13: Distinguish clearly between Absorption and Assimilation.

- **Absorption:** It is the physical movement of simple, digested nutrients across the microscopic membrane walls of the intestinal villi.
- **Absorption:** It transfers these nutrients directly into the local bloodstream or the lymphatic lacteal system.
- **Assimilation:** It occurs when individual body cells take in these absorbed nutrients from the blood and utilize them for energy or growth.

Q14: How does the single-cell-thick epithelium of a villus adapt for its function?

The single layer of simple columnar epithelium minimizes the physical thickness of the barrier that nutrients must cross. According to diffusion laws, reducing the physical distance dramatically increases the overall speed and efficiency of transport. It allows rapid and continuous passage of glucose, amino acids, and minerals into underlying blood vessels via diffusion.

Q15: How does the liver prevent the body from suffering from ammonia poisoning?

When the body metabolizes excess amino acids, it removes their amino groups through a process called deamination. This metabolic breakdown produces highly toxic ammonia, which can damage cells and the brain if it accumulates. The liver instantly traps this dangerous ammonia and converts it through chemical cycles into much less toxic urea for kidney excretion.

Q16: What metabolic process occurs in the liver when blood glucose levels drop drastically during fasting?

- **Glycogenolysis:** The liver breaks down its stored reserves of glycogen back into pure glucose to release into the bloodstream.
- **Gluconeogenesis:** If fasting continues and glycogen runs out, the liver activates an alternative metabolic pathway to synthesize new glucose.
- **Gluconeogenesis:** It chemically transforms non-carbohydrate sources like glycerol, lactic acid, and amino acids into functional glucose.

Q17: Why is the liver considered a recycling center for red blood cells?

Red blood cells only live for about 120 days before their cell membranes become worn out and fragile. Specialized phagocytic cells in the liver track down, capture, and break down these old, non-functional red blood cells. The liver dismantles the hemoglobin molecules to salvage and store valuable iron ions, which are later reused to produce new cells.

Q18: Why does severe diarrhea lead to a medical emergency if left untreated?

Hyperactive peristalsis forces liquid waste matter through the large intestine too quickly for proper water reabsorption to occur. This results in a massive, rapid, and unchecked loss of water along with critical body electrolytes like sodium and potassium. The resulting acute dehydration causes blood volume to plunge drastically, leading to organ failure and fatal hypovolemic shock.

Q19: Explain how a lack of dietary fiber (roughage) directly causes constipation.

Dietary fiber is indigestible plant cellulose that acts like a sponge, holding onto water and adding vital physical bulk to stool. Without fiber, fecal matter becomes extremely small, highly compressed, dense, and dry within the large intestine. This lack of physical bulk fails to stretch the intestinal walls, slowing down peristalsis and allowing the colon to over-absorb remaining water.

Q20: Why can emotional stress or heavy smoking lead to the formation of gastric ulcers?

High emotional stress triggers the autonomic nervous system to hyper-secrete corrosive gastric hydrochloric acid into the stomach. Concurrently, nicotine from cigarette smoking restricts blood flow to the stomach lining, slowing down the regeneration of protective cells. This combination severely degrades the protective mucus barrier, allowing excess acid to burn, corrode, and erode the stomach wall tissue.

LONG QUESTIONS DIGESTIVE SYSTEM

Q1: Describe the structures of the main regions of the alimentary canal.

The alimentary canal is also known as the gut or digestive tract, or gastrointestinal tract. It is a continuous tube. The tube is about 9 meters long in adults. It is specialized at various points along its length. Each region is designed to carry out a different function in the overall process of digestion, absorption, and egestion. The alimentary canal begins from the mouth and ends at the anus. It is lined internally by a mucous membrane.

- **Mouth**

It is the external opening that leads into the oral cavity or mouth cavity.

- **Oral Cavity**

Around the sides and front of the oral cavity are the upper and lower jaws. The upper jaw is fixed while the lower is movable. The oral cavity contains teeth and the tongue.

- **Pharynx**

It leads from the mouth and nasal cavity to the oesophagus and to the trachea. The pharynx is a common passage for food and air.

- **Oesophagus**

It is a muscular tube, about 25 cm long. It leads from the pharynx to the stomach. In the oesophagus, no digestion takes place.

- **Stomach**

The stomach is the dilated part of the digestive tube. It is roughly J-shaped and somewhat transversely placed on the left of the abdomen just beneath the diaphragm. It has two openings, which are guarded by valves. The opening between the oesophagus and the stomach is called the cardiac opening, which is guarded by the cardiac sphincter. The region of the stomach around the cardiac opening is the cardiac end. The opening of the stomach into the duodenum is called the pyloric opening, which is guarded by muscles called the pyloric sphincter.

- **Small Intestine**

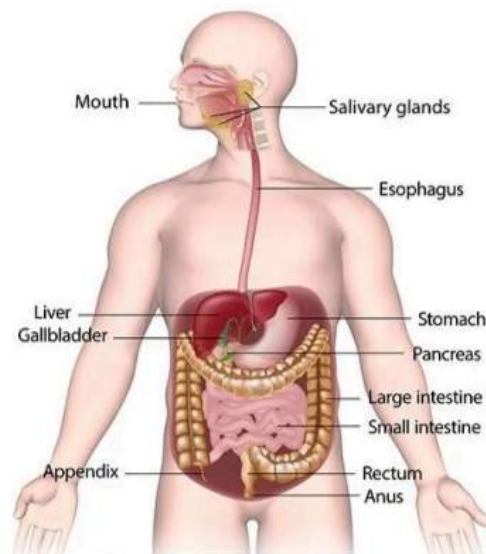
The small intestine is small in diameter. It is divided into three parts: duodenum, jejunum, and ileum. The duodenum is about 20-25 cm long. The bile duct from the gall bladder, the hepatic duct from the liver, and the pancreatic duct from the pancreas empties its secretions into the duodenum. The jejunum is about 2.5 meters long. The ileum is about 4 meters long.

- **Large Intestine**

The large intestine is a tube leading from the small intestine to the anus. It is 1.5 meters in length. It is larger than the small intestine in diameter. The large intestine is made up of three parts: the colon, the rectum, and the anus. The colon

can also be divided into parts. The caecum is the first part of the colon. The ileum enters it from the side. The caecum has a pouch-like end to its base appendix is attached, which is 10 cm long.

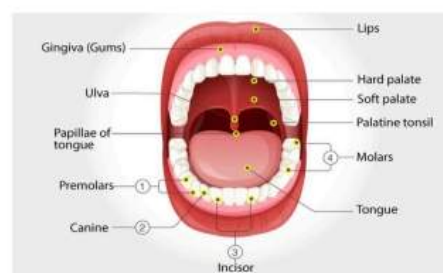
The rest of the colon is divided into segments: the ascending colon (right), the transverse colon (across), the descending colon (left), and the sigmoid colon, which connects to the rectum. The rectum is a straight, 8-inch chamber that connects the large intestine to the anus. The rectum temporarily stores the feces. The anus is the last part of the digestive tract. It is a 2-inch-long canal. The anus is surrounded by sphincter muscles that are important in allowing control of stool.



Q2: Describe the structures of the associated organs of the alimentary canal.

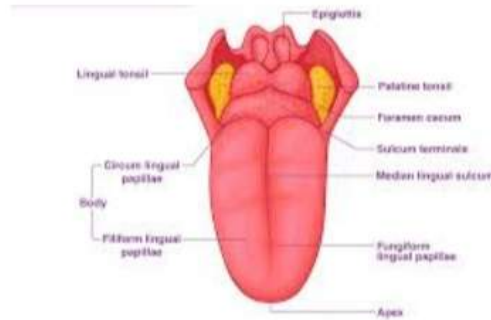
Associated organs include teeth, tongue, salivary glands, liver, gallbladder, and pancreas.

1. Teeth :The four types of teeth are incisors, canines, premolars, and molars. Incisors bite into food and cut it into smaller pieces. Canines are used to tear food. Premolars help to chew and grind up food. Molars chew food and grind it up.



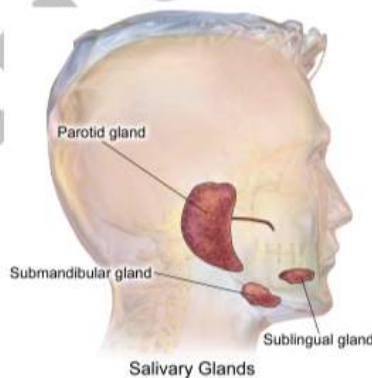
2. Tongue

The tongue is a muscular organ in the mouth cavity. It is a digestive organ. The tongue moves food around the mouth to help mastication and swallowing. It also helps in chewing, speaking, and breathing. The tongue has taste buds.



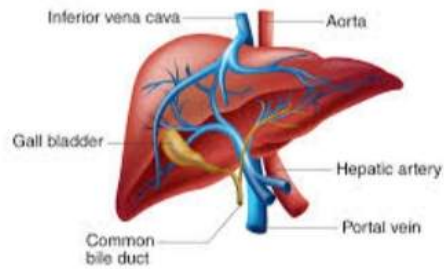
3. Salivary glands

The salivary glands produce saliva. The three major pairs of salivary glands are the sublingual glands, the submandibular glands, and the parotid glands. Sublingual glands are below either side of the tongue, under the floor of the mouth. Submandibular glands are located below the jaw. Like the sublingual glands, the saliva produced in the submandibular glands enters the mouth from under the tongue. Parotid glands are just in front of the ears. The saliva produced by the parotid glands enters the mouth from small ducts near the upper molars.



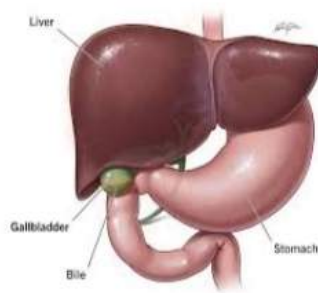
4. Liver

The human liver is an organ and gland in the human body. It is spongy, wedge-shaped, and reddish-brown in color. The liver is located on the right side of the body, under the ribs. The liver secretes bile. The liver has two main parts: the larger right lobe and the smaller left lobe. The lobes contain many blood vessels. Blood travels through the liver. The lobes also contain thousands of small lobes, i.e., lobules. These lobules connect with many bile ducts. These ducts transport bile from the liver to the duodenum.



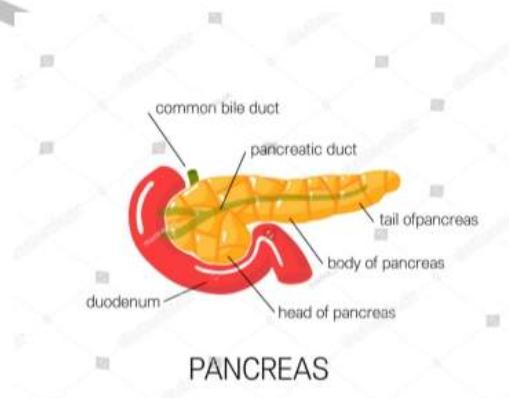
5. Gallbladder

The gallbladder is just below the liver. Bile is stored in the gallbladder and flows through the cystic duct and the common bile duct into the small intestine when food is being digested.



6. Pancreas

The pancreas is behind the stomach and in front of the vertebral column. It is surrounded by the gallbladder, liver, and spleen. The head of the pancreas is on the right side of the body. It is tucked beside the curve of the duodenum. The tail of the pancreas extends over to the left side of the body, near the spleen.

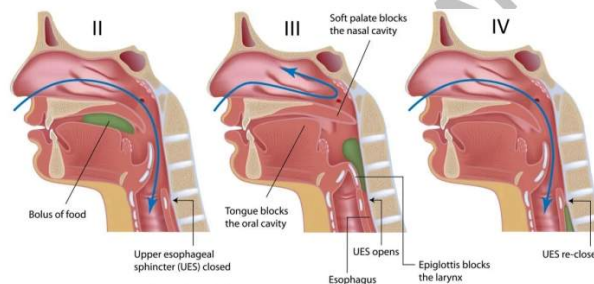


**Q 3: Describe the process of swallowing food in humans.**

In swallowing, the following actions take place:

- (a) The tongue presses upwards and back against the roof of the mouth, forcing the bolus to the back of the mouth.
- (b) The soft palate closes the nasal cavity at the back.
- (c) The larynx cartilage around the top of the trachea is pulled upwards so that the glottis (opening of the trachea) lies under the back of the tongue.
- (d) The epiglottis (a flap of cartilage) helps to prevent food from entering the glottis instead of the oesophagus.

The beginning of the swallowing action is voluntary. But once the food reaches the back of the mouth, swallowing becomes an automatic or involuntary action. The food is forced into and down the oesophagus by peristalsis.

**Q 4: Describe the process of peristalsis in man with a diagram.**

The alimentary canal has layers of muscle in its wall that push the ingested food forward. A contraction of circular muscles just behind the bolus pushes the food forward into the next region, where longitudinal muscles are contracted, and the oesophagus becomes wide. This wave of contraction is called peristalsis. It is the rhythmic wave of muscular contraction and relaxation in the wall of the alimentary canal that causes the food to move through the alimentary canal.

(Note: In your handwritten exam, you should sketch a simple diagram showing a wave of contraction behind the food bolus and an area of relaxation ahead of it, as shown below)

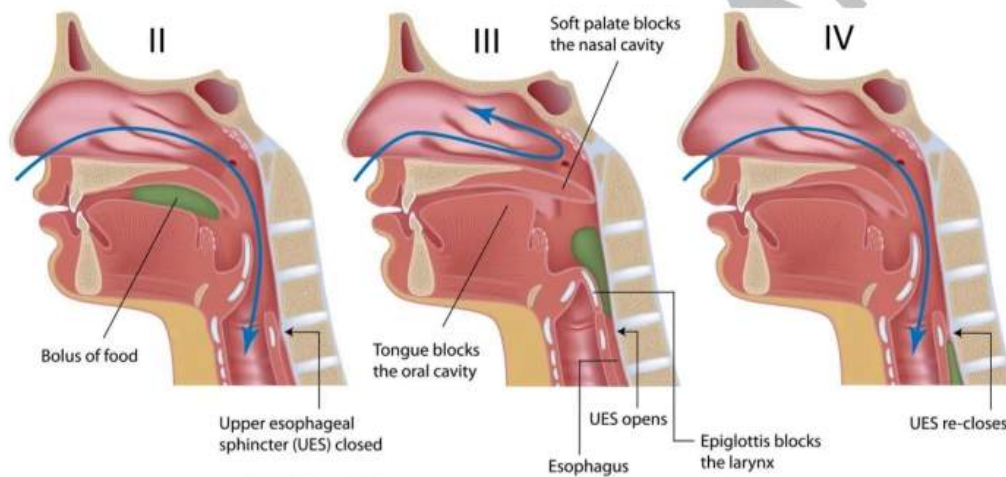


Q 3: Describe the process of swallowing food in humans.

In swallowing, the following actions take place:

- (a) The tongue presses upwards and back against the roof of the mouth, forcing the bolus to the back of the mouth.
- (b) The soft palate closes the nasal cavity at the back.
- (c) The larynx cartilage around the top of the trachea is pulled upwards so that the glottis (opening of the trachea) lies under the back of the tongue.
- (d) The epiglottis (a flap of cartilage) helps to prevent food from entering the glottis instead of the oesophagus.

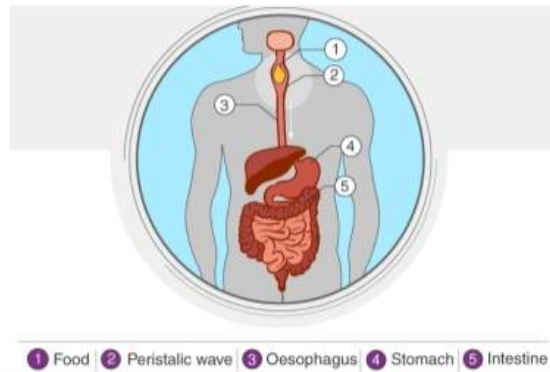
The beginning of the swallowing action is voluntary. But once the food reaches the back of the mouth, swallowing becomes an automatic or involuntary action. The food is forced into and down the oesophagus by peristalsis.



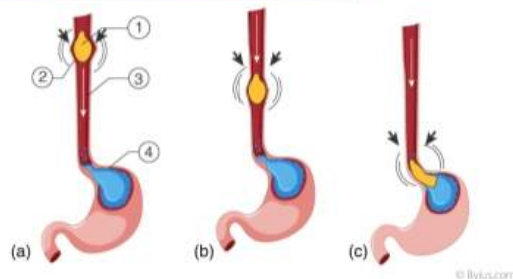
Q 4: Describe the process of peristalsis in man with a diagram.

The alimentary canal has layers of muscle in its wall that push the ingested food forward. A contraction of circular muscles just behind the bolus pushes the food forward into the next region, where longitudinal muscles are contracted, and the oesophagus becomes wide. This wave of contraction is called peristalsis. It is the rhythmic wave of muscular contraction and relaxation in the wall of the alimentary canal that causes the food to move through the alimentary canal.

(Note: In your handwritten exam, you should sketch a simple diagram showing a wave of contraction behind the food bolus and an area of relaxation ahead of it, as shown below)



ESOPHAGEAL PERISTALSIS



Q 5: Describe the process of digestion in the stomach.

On the peristaltic reflex, the cardiac sphincter relaxes to allow food in the stomach. Gastric Juice: The food in the stomach stimulates the gastric glands to secrete gastric juice.

It consists of mucous, HCl, pepsin, and rennin.

- **Mucus:** Mucus forms a coating over the stomach. It prevents the digestion of the stomach wall by the enzyme pepsin.
- **Hydrochloric Acid:** HCl performs the following functions:
 - (a) Stops the action of salivary amylase.
 - (b) It converts the inactive form of the enzyme (pepsinogen) to the active form (pepsin). It provides an acidic medium suitable for the action of enzymes.
 - (c) It kills many microorganisms.
- **Pepsin:** It is secreted as inactive pepsinogen. HCl or already activated pepsin converts pepsinogen to pepsin. Pepsin digests proteins into polypeptides (chains of many amino acids) and peptides (chains of a few amino acids).
- **Rennin:** Rennin is secreted only in infants. It helps in the digestion of milk protein. By the churning action of the stomach wall, the food gets mixed into a soupy mixture called chyme. When the chyme reaches a certain degree of acidity, the pyloric sphincter relaxes, and a little food passes into the duodenum.

Q6: How does the process of digestion take place in the small intestine?

Starch, proteins, and fats are all enzymatically broken down in the small intestine. When food enters the duodenum, the chyme stimulates:

- (a) The gall bladder is to release bile.
 - (b) The pancreas secretes pancreatic juice.
 - (c) The intestinal glands secrete intestinal juice.
- All these secretions are alkaline and neutralize the acidic chyme.

The pancreatic and intestinal enzymes can work only in this alkaline medium.

- **Bile :**

Bile is a greenish-yellow liquid manufactured in the liver. Bile is stored in the gallbladder and is released as soon as food enters the duodenum. Bile emulsifies fats, causing them to break down into small droplets called an emulsion. Emulsification provides relatively large surface areas of lipid for the action of lipase enzymes.

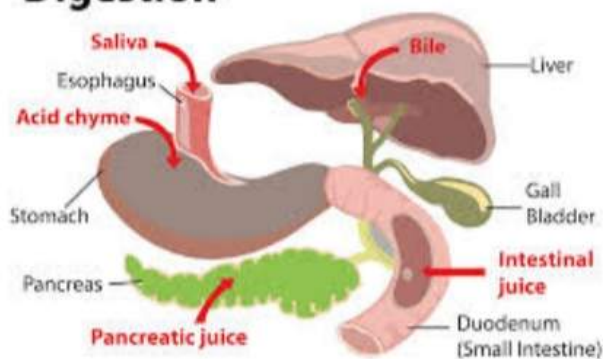
- **Pancreatic Juice**

The secretion of the pancreas is called pancreatic juice. It contains enzymes amylase, lipase, and trypsinogen. Pancreatic amylase is the starch-digesting enzyme. It converts polysaccharides to maltose and even to glucose. Pancreatic lipase converts neutral fats (i.e., triglycerides) into glycerol and fatty acids. Trypsin converts proteins into polypeptides.

- **Digestion in Jejunum and Ileum**

The lining of the jejunum and ileum secretes several enzymes, such as erepsin, which converts peptides into amino acids. It also secretes enterokinase, which activates pancreatic trypsinogen into its active form, trypsin. Lactase converts lactose to glucose. Maltase converts maltose to glucose, etc. In the ileum, the digestion process is complete.

Digestion



Q 8: State the role of the liver.

- 1) The liver cells manufacture bile.
- 2) An excess quantity of glucose is stored by the liver as glycogen.
- 3) The liver converts glycerol and amino acids to glucose molecules.
- 4) The liver removes the amino group from amino acids. It is called deamination.
- 5) Ammonia is converted into urea in the liver.
- 6) The majority of the plasma proteins are synthesized in the liver.
- 7) It destroys old red blood cells.
- 8) It stores fat-soluble vitamins (A, D, E, and K) and mineral ions such as iron.

Q 9: Write the symptoms, cause, treatment, and prevention of the following disorder of the digestive system.

a. Diarrhoea

- **Symptoms:** The stools become watery. There is an increased frequency of defecation.
- **Causes:** The major causes are infection of the lower GI tract and nervous stimulation. In case of infection, such as food poisoning caused by eating contaminated food, the intestinal wall becomes irritated and peristalsis increases.
- **Treatment:** Diarrhoea may result in dehydration, so ORS (oral rehydration solution) or a mixture of salt and sugar solution is given to the patient.
- **Prevention:** Always consume clean water, eat freshly prepared food, and maintain general hygiene.

b. Constipation

- **Symptoms:** Include a decrease in frequency of daily stools or difficulty in defecation, which may result in abdominal pain.
- **Causes:** The faeces move too slowly along the large intestine. As a result, more water is absorbed from the faeces than usual. The faeces become so hard that defecation becomes difficult and painful. Some of the most common causes are: medications, lack of exercise, not enough liquids, and not enough fiber in the diet, such as fruit, vegetables, and cereals, a change in your routine or lifestyle, such as a change in your eating habits.
- **Treatment:** For the treatment of constipation, a laxative is taken to make the faeces soft for easy discharge. Constipation is called the mother of all diseases.
- **Prevention:** As a preventive measure, one should take plenty of water or fluids and fibrous food, fruits, and vegetables, etc.

c. Ulcer

- **Symptoms:** The main symptom of an ulcer is stomach pain.

- **Causes:** An ulcer is an open sore in the wall of the gut caused by the gradual disintegration of the tissue. It may be a gastric (stomach) ulcer or a peptic (duodenal) ulcer. The causes of ulcers are acidity, smoking, and bacterial infection.
- **Treatment:** Doctors usually treat ulcers with medications to lower stomach acid or specific antibiotics to cure bacterial infections.
- **Prevention:** For the prevention of ulcers, (a) stop smoking, (b) avoid spicy food and food containing acids, (c) avoid stress.

MA'AM SHABANA